

EXAM 7 – SPRING 2015

19. (1.25 points)

A monoline property insurance company writing business exclusively in the U.S. requires catastrophe reinsurance in order to remain within acceptable levels of risk tolerance. The company is considering two different reinsurance options from reinsurers willing to write 100% of the risk for the same price.

- Reinsurer A has \$10 billion of surplus, writes primarily U.S. property reinsurance and has the highest possible financial strength ratings.
- Reinsurer B has \$10 billion of surplus, writes primarily casualty reinsurance and has the highest possible financial strength ratings.

a. (0.75 point)

Explain how the risk level differs between the two options and how that might impact the required surplus as measured by the company's capital model.

b. (0.5 point)

Explain why placing 100% of the coverage with one reinsurer may lead to a suboptimal use of capital for the company.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 19	
TOTAL POINT VALUE: 1.25	LEARNING OBJECTIVE: C1: Demonstrate how insurance and financial risk can be analyzed quantitatively. C8: Evaluate best practices in risk measurement, modeling, and management of various financial and non-financial risks faced by an entity.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> While reinsurer A has a better knowledge of property reinsurance, it is also much more vulnerable to catastrophe risk as it could affect a large portion of its portfolio at once. This would not be the case for reinsurer B as geographical and line of business diversification would lessen the impact of a catastrophe (even though casualty reinsurance would still be highly correlated in the event of a large catastrophe). With these considerations in mind reinsurer A would require a higher reinsurance risk margin for the primary insurer's capital requirements. <i>Sample Answer 2</i> Both are similar financially but are in different industries. Reinsurer B will be diversified since a cat event will impact both a primary property insurer and property reinsurer. The risk measure should be lower using company B over company A. Same with the required surplus.	
Part b: 0.5 point	
<i>Sample Answer 1</i> This increases counterparty risk and the amount of capital that must be held to cover that risk. Separating the reinsurance agreement could help lower the capital requirement for that risk as it reduces the insurer's exposure to its reinsurer's failure. <i>Sample Answer 2</i> Assuming losses between these reinsurers are not completely dependent, it may be more optimal to spread coverage between multiple reinsurers to lower credit risk. This would allow the insurer to hold less capital for credit risk.	
EXAMINERS' REPORT	
Candidates were expected to know some basic concepts and best practices in ERM models. Candidates generally performed well on this question and were able to recognize the major issues presented by the problem. Where candidates struggled was with more subtle parts of the question.	
Part a	
Most candidates recognized and explicitly stated that the correlation between the underwriting exposures for the primary insurer and reinsurer A would lead to the capital model indicating more surplus should be held if the reinsurance coverage were placed with reinsurer A. Candidates struggled to make explicit that this correlation is stronger in the far right tail of the distribution with which capital models are generally more concerned. Given that the scenario presented involved catastrophe reinsurance, this was implicit in some responses, but the tail of the	

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distribution for an insurer and reinsurer do not always overlap so well.

One mistake several candidates made was in mentioning underwriting guidance, which is given as a reason for purchasing reinsurance in the Patrik paper. While this is in some cases useful, if it is at all quantified in the capital model, it is far lower in magnitude than the correlation of underwriting exposures in the tail. Several candidates mentioned that property reinsurance was more likely to lead to a generally quicker bankruptcy than latent claims which, while true, ignores the more salient point of the correlation in tail exposures.

Part b

In part b, most candidates recognized that placing coverage among multiple reinsurers *may* lead to some net benefit. Diversifying coverage is less risky in parts of the model, but given the issues in part a, placing any reinsurance with reinsurer A will lead to higher indicated surplus in other parts of the model. Whether or not this is a net benefit overall depends on the model. Several candidates stated that diversification was always unconditionally good, which is incorrect. Some candidates appeared to misinterpret "100% of the coverage" as fully ceded property premium and gave a response relating to return on capital.